# CONDITIONS ON A POINT OF AN ALGEBRAIC SPACE OVER A FIELD

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*Editorial note.* No omission markers in-span.

*Allowed standard prerequisites (editorial).*

- Fibres over fields. lemma-decent-finite-fibre (carded L26740). The fibre is a finite union of field spectra.
- Constructible points over fields. spaces-properties lemma constructible-points (cross-chapter, named). The scheme case over a field.

*Source reference.* Original reference 00OU: focused-seeds/geometry/S9-decent-conditions-on-point-on-space-over-field/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

*Source reference.* Original lemma 0A4J: sources/stacks-decent-spaces/pinned-original-source.tex, source lines 2945–2945. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

*Source reference.* Original reference 01TH: focused-seeds/geometry/S9-decent-conditions-on-point-on-space-over-field/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

*Source reference.* Original reference 04NM: focused-seeds/geometry/S9-decent-conditions-on-point-on-space-over-field/ref-tags.txt, source lines 7–7. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

*Source reference.* Original reference 04NW: focused-seeds/geometry/S9-decent-conditions-on-point-on-space-over-field/ref-tags.txt, source lines 8–8. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

*Source reference.* Original reference 06LS: focused-seeds/geometry/S9-decent-conditions-on-point-on-space-over-field/ref-tags.txt, source lines 9–9. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction

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**Lemma 0.1.** *Let $X$ be an algebraic space locally of finite type over a field $k$. Let $x \in |X|$. Consider the conditions*

(1) $\dim_x(|X|) = 0$,
(2) *$x$ is closed in $|X|$ and if $x' \rightsquigarrow x$ in $|X|$ then $x' = x$,*
(3) *$x$ is an isolated point of $|X|$,*
(4) $\dim_x(X) = 0$,
(5) $X \to \mathrm{Spec}(k)$ *is quasi-finite at $x$.*

*Then (2), (3), (4), and (5) are equivalent. If $X$ is decent, then (1) is equivalent to the others.*

*Proof.* Parts (4) and (5) are equivalent for example by Morphisms of Spaces, Lemmas 04NW and 06LS.

Let $U \to X$ be an étale morphism where $U$ is an affine scheme and let $u \in U$ be a point mapping to $x$. Moreover, if $x$ is a closed point, e.g., in case (2) or (3), then we may and do assume that $u$ is a closed point. Observe that $\dim_u(U) = \dim_x(X)$ by definition and that this is equal to $\dim(\mathcal{O}_{U,u})$ if $u$ is a closed point, see Algebra, Lemma 00OU.

If $\dim_x(X) > 0$ and $u$ is closed, by the arguments above we can choose a nontrivial specialization $u' \rightsquigarrow u$ in $U$. Then the transcendence degree of $\kappa(u')$ over $k$ exceeds the transcendence degree of $\kappa(u)$ over $k$. It follows that the images $x$ and $x'$ in $X$ are distinct, because the transcendence degree of $x/k$ and $x'/k$ are well defined, see Morphisms of Spaces, Definition 04NM. This applies in particular in cases (2) and (3) and we conclude that (2) and (3) imply (4).

Conversely, if $X \to \mathrm{Spec}(k)$ is locally quasi-finite at $x$, then $U \to \mathrm{Spec}(k)$ is locally quasi-finite at $u$, hence $u$ is an isolated point of $U$ (Morphisms, Lemma 01TH). It follows that (5) implies (2) and (3) as $|U| \to |X|$ is continuous and open.

Assume $X$ is decent and (1) holds. Then $\dim_x(X) = \dim_x(|X|)$ by Lemma 0A4J and the proof is complete. □